High Harmonic Generation with Plasmonic Enhancements

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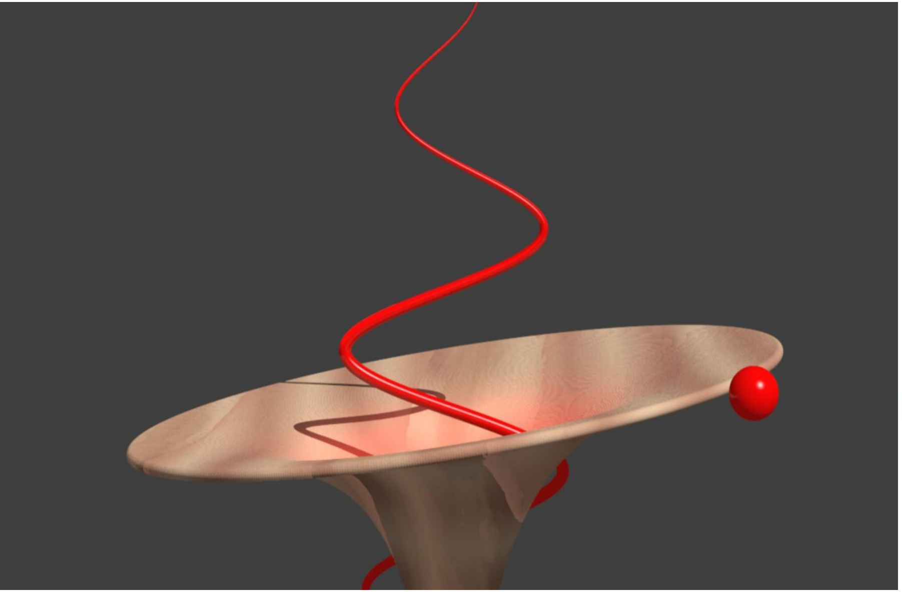
Introduction to Attoscience



- One attosecond = 1x10^-18 seconds
- Strong field = coulomb potential

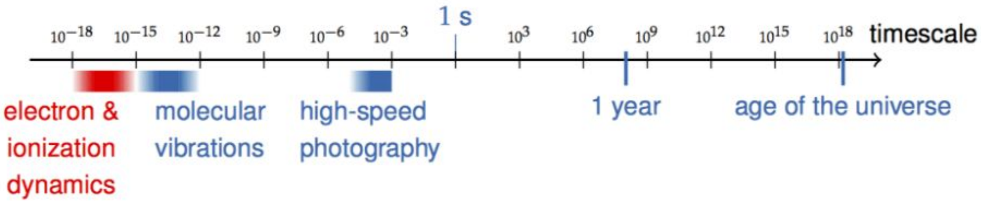


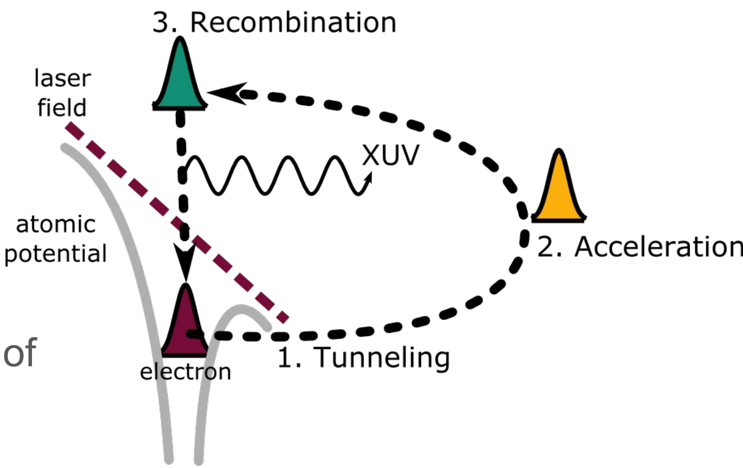
Image By A C Bray and C Hofmann

Systems of Interest

- Analysis of the impact of lasers and their strong fields on electrons bound to their parent ions

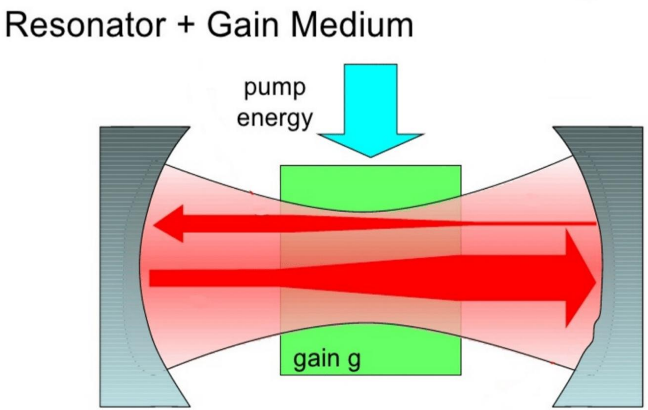
- 3 step model: **Tunneling, Acceleration, Recombination**

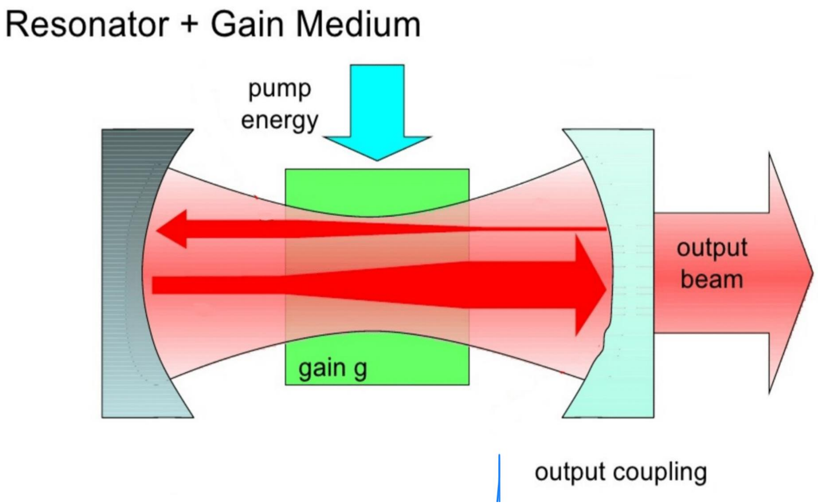
- The results we obtain will differ depending on the properties of the laser.



What is a laser?

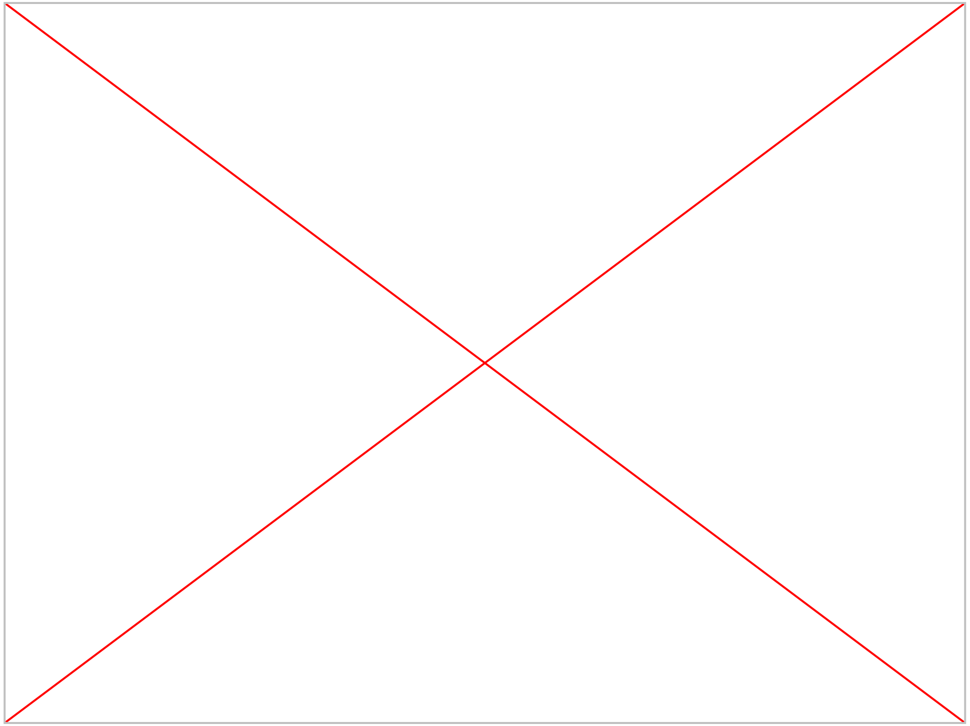
Stimulates atoms to emit light at different wavelength





Types of polarisation

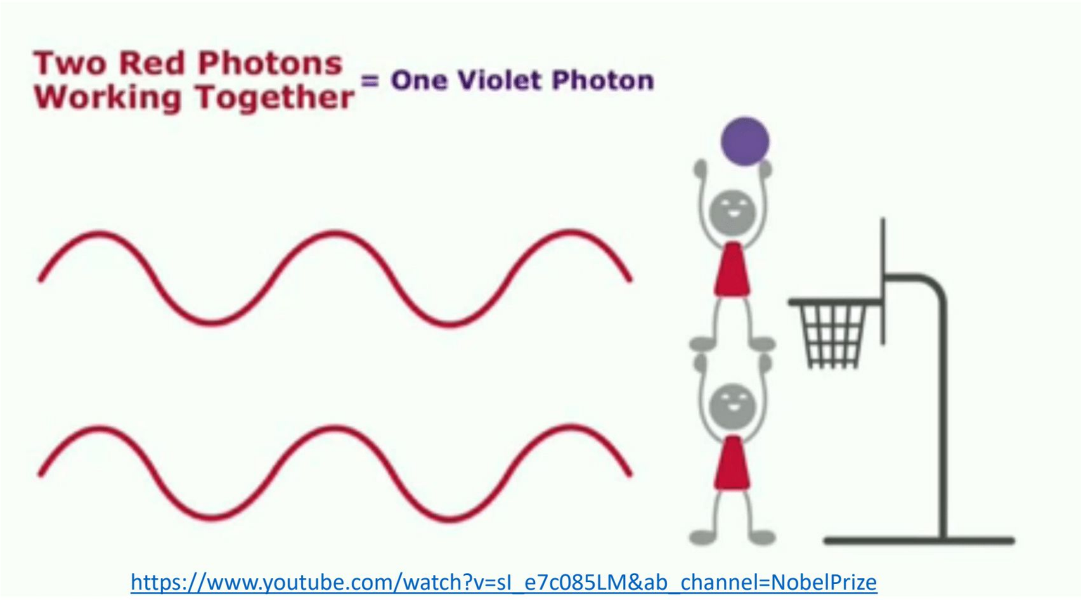
- Linear
- Circular
- Elliptical



Introduction to Polarization-https://www.edmundoptics.co.uk/knowledge-center/application-notes/optics/introduction-to-polarization/

Motivation for XUV

- Input IR to XUV
- Attoseconds
  vs.
  femtoseconds



Forces

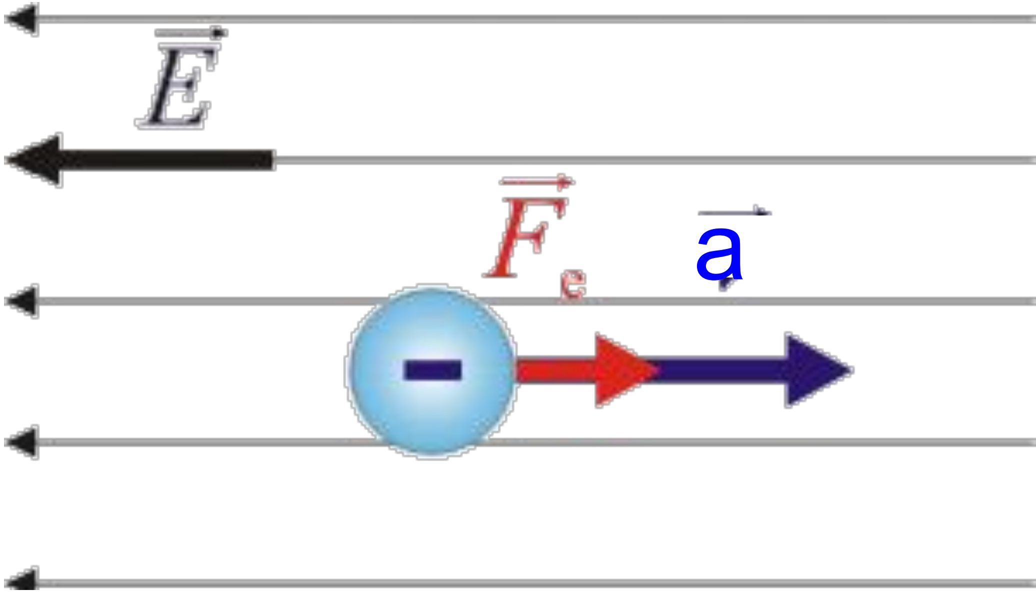
Force and **electron motion**

Formula to use: **F = ma**

In this case, we will make the **mass of the electron** to have a unit mass of “**1**”, therefore **F = a**

The **electric field** that interacts with the electron will experience a **force that’s in opposite in direction** with the **electric field, E(t)**

So E(t) = -F, E(t) = -a.



Trajectory

- Returning near ion for **HHG-recombining.**
- How electrons recombine?
- Effect of **laser field** on **electrons leaving parent ion.**
- What is a **trajectory?**
- Many electrons have different **trajectories. Long** and **short?**

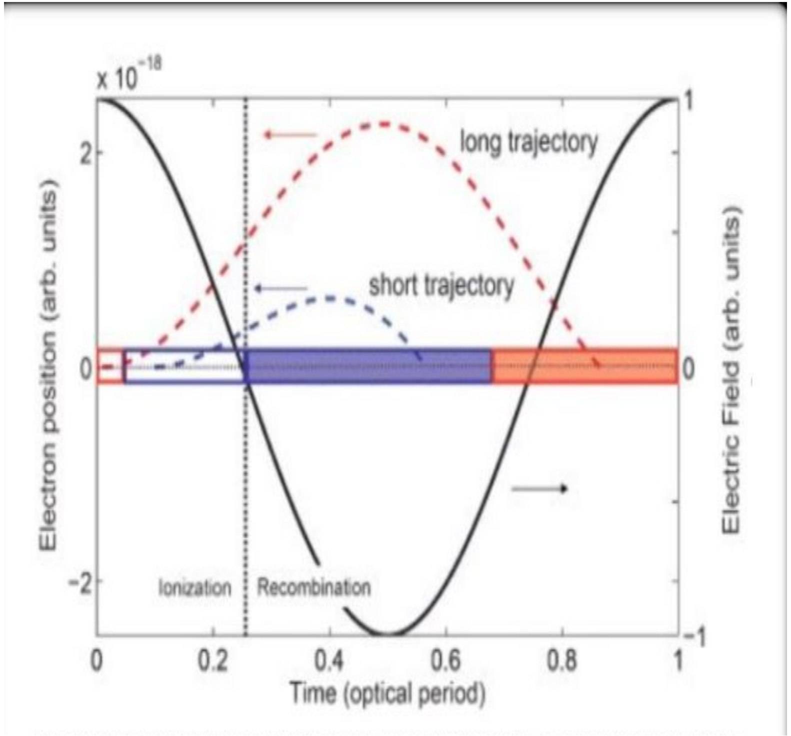
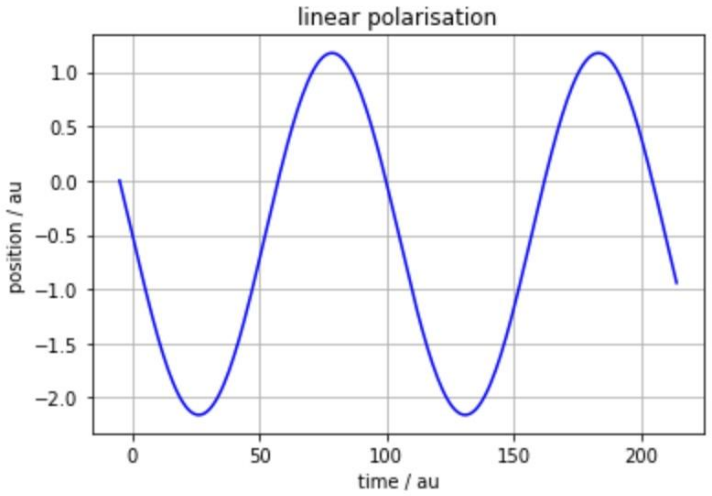


Image by Deutsche Physikalische Gesellschaft

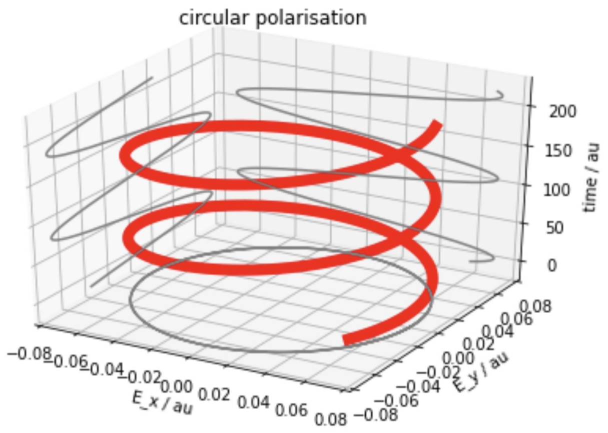
Linear Polarisation



During Linear polarisation,electrons move back and forth in one line and in any direction

The electron does recombine.

Circular Polarisation



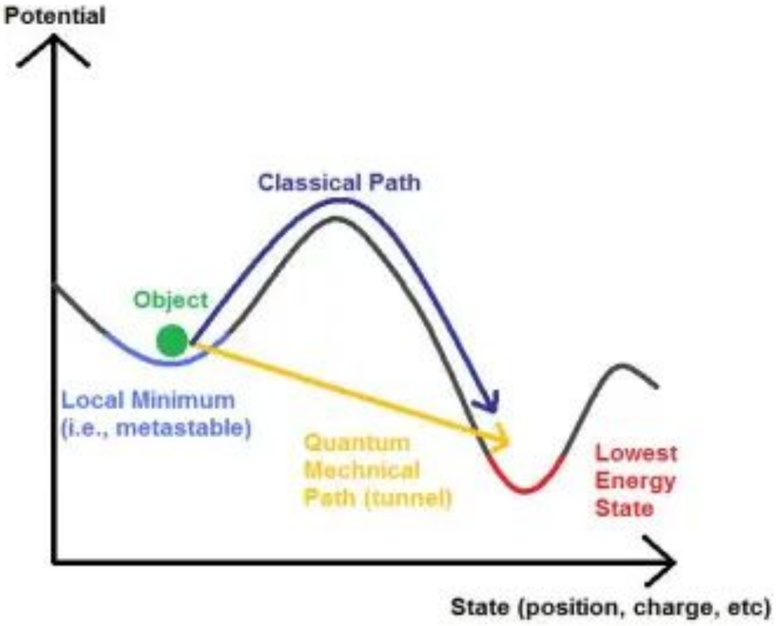
The resulting electric field rotates in a circle around the direction of propagation.

During circular polarisation, the electron does not recombine with the parent ion.

Quantum Tunnelling

Takes effect when Electrons move through **barriers** you wouldn't expect them to.

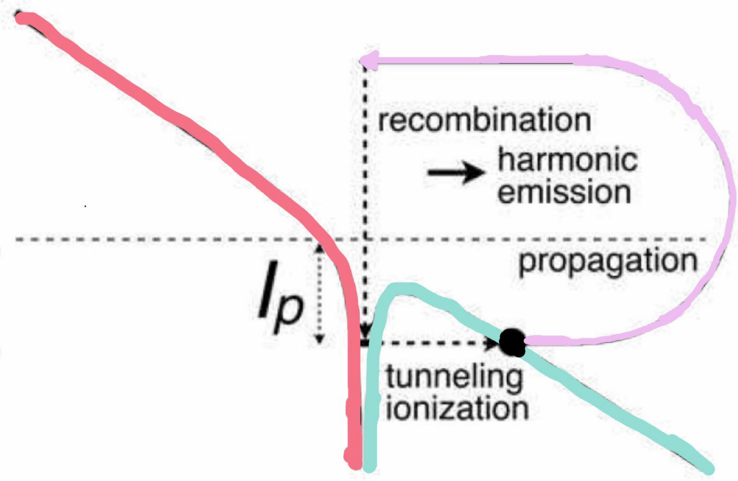
A strong electric field disrupts the coulomb potential of an electron. An electron is **accelerated** direction of the **trajectory** is changed (The **field** begins to **oscillate**).



Link: https://bigthink.com/technology-innovation/physicists-leverage-quantum-tunneling-to-collect-energy-from-earths-heat/

The Three Step Model

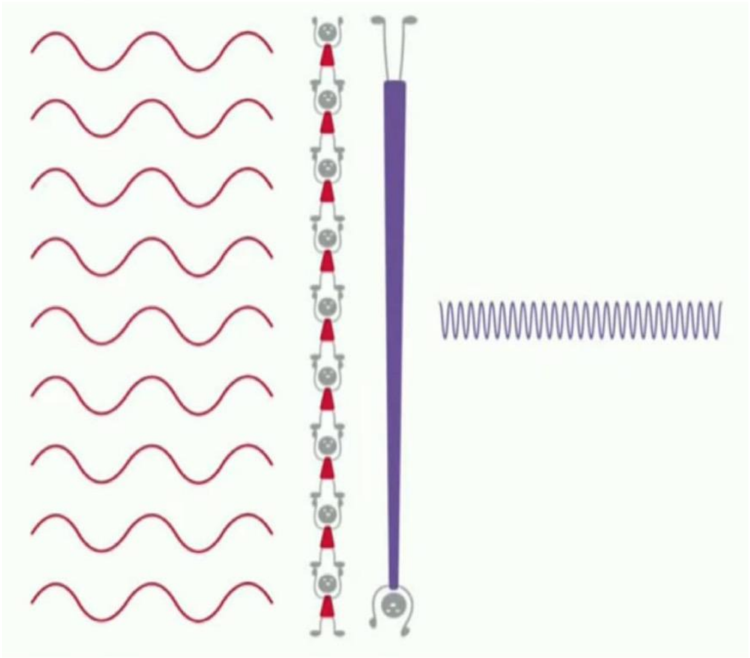
1. Tunnel Ionisation
   - An electron leaves the target atom/molecule through the process of quantum tunneling and is **accelerated** by the laser field.
2. Propagation/Acceleration
   - The released electron **propagates in the continuum**, on a trajectory outside the target atom/molecule until it travels close to the Coulomb potential of its parent ion
3. Recombination
   - The electron returns to its parent ion with more energy

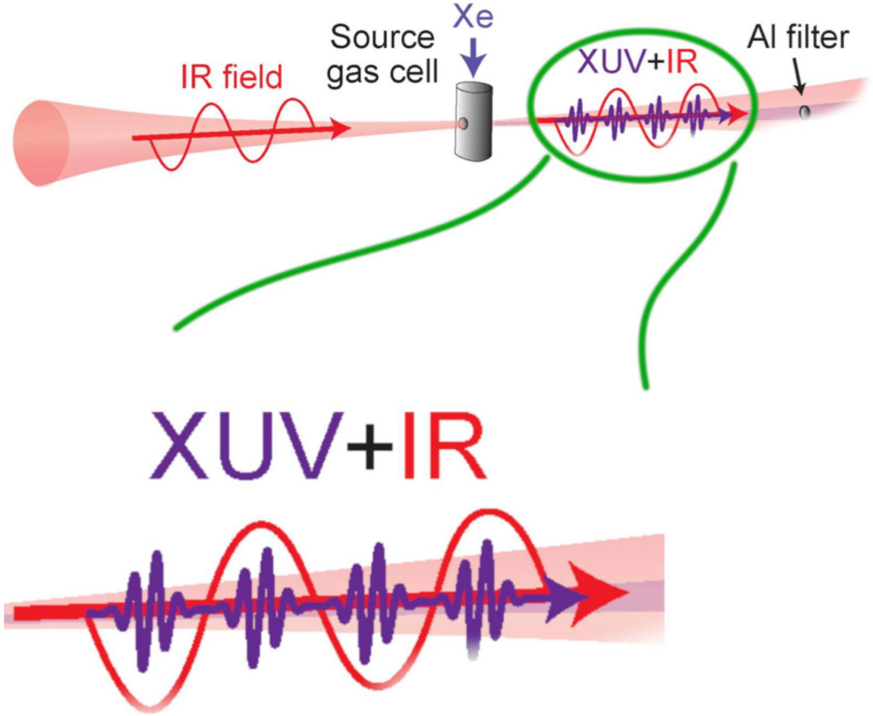


Reference: DOI:10.5772/7961

High Harmonic Generation

Energy of electron at recombination ≅ Energy of XUV Pulse





What is the optical model?

- Our **optical model** is a **superposition** of **two circularly polarised EM waves**
- In what is called the homogenous setup One EM wave **rotates in opposite direction 2x frequency** (see **FIG 1**) of the other

- Relative **rotational frequency is 3** (1 cw, -2 ccw, 1-(-2)=3)

- Leads to **trifold petal pattern** (see **FIG 1**) with an **order** of rotational symmetry of **3** with a maxima every ⅓ of a rotation (optical period of 1/3)

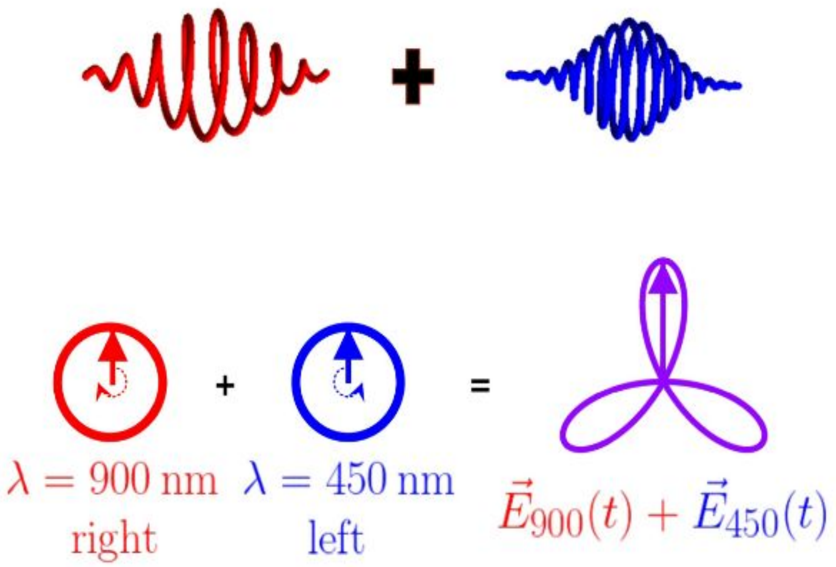


FIG 1

Why do we use this optical model?

- Used because its super easy to modify shape of wave by slight deviations from the **homogenous case.**

- The **linear** model is more **limited** in the waveforms produced, because it is more **difficult to manipulate XUV pulses.**

- But expectedly in the **homogenous** case, (see **FIG 2**) we see a regular **sinusoidal shape** representing the **maximas passing through the atom.**

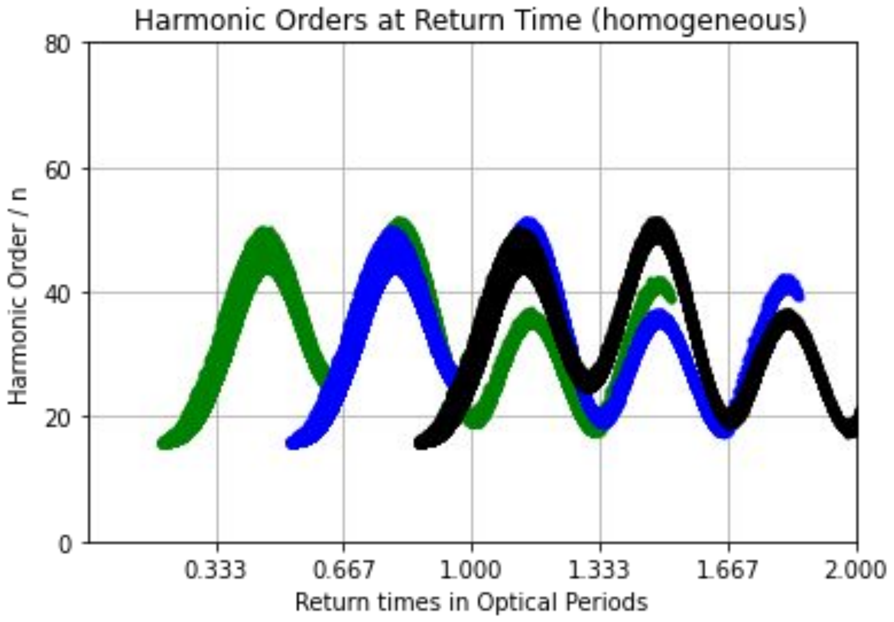
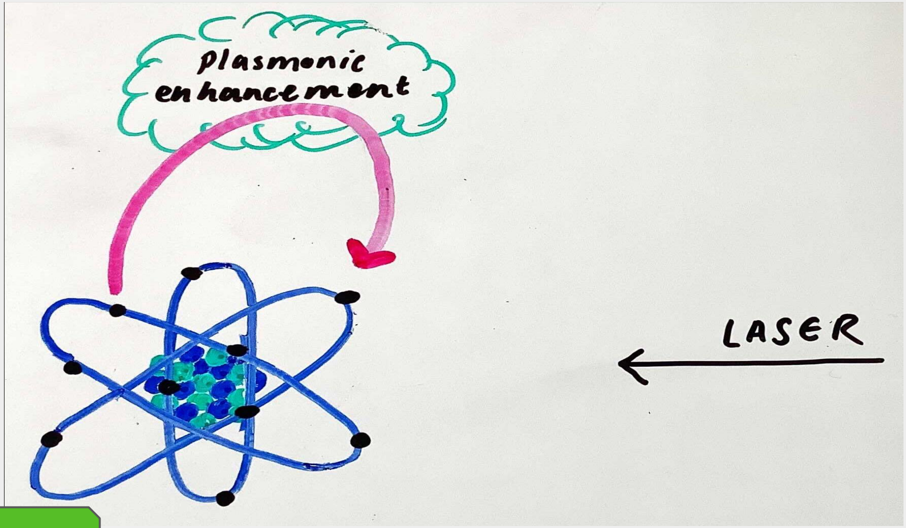


FIG 2

Plasmonic enhancement:

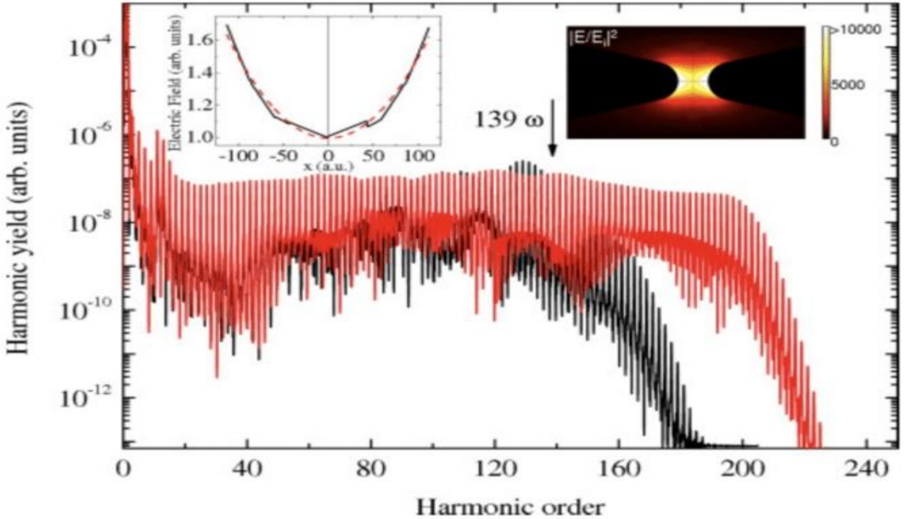
So far, we have a scenario in which a laser fires at a neon atom causing an electron to be **ejected** and undergo the **three step model**.

However, we could add **plasmonic enhancement.**



- Nanosphere located at x = 297 au
- Electric field strength = length of arrow

Our parameters with plasmonic enhancement



argon, $\lambda = 1.8\ \mu m$, $I = 8 \cdot 10^{13}\ W/cm^2$, TDSE

$\Rightarrow$ higher energy spectrum

M. F. Ciappina, et al., *Optics Express*, **20**, 26261, 2012.

Using plasmonic enhancements, we are able to reach a higher harmonic order

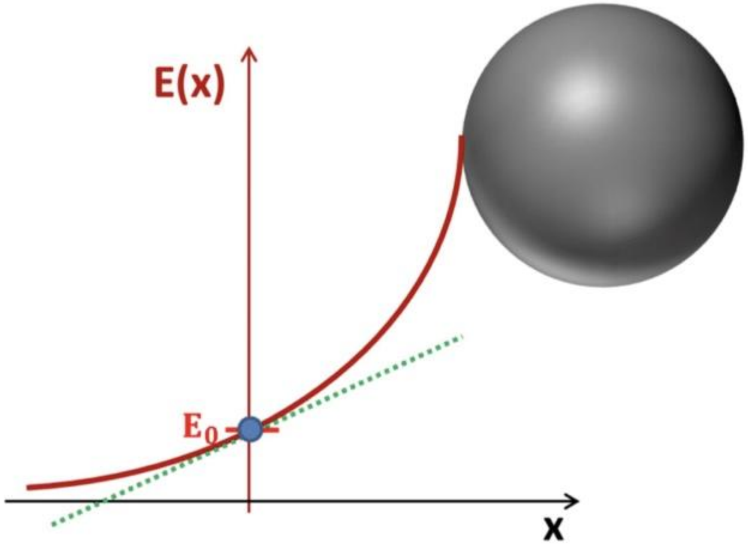
This enables us to investigate more values and make more observations on them

Key:

- **Plasmonic enhancement**
- **No plasmonic enhancement**

Linear vs cubic model

Linear model	Cubic model
• Only works for a **small range** of **travel distance** for the electrons • Not a good mathematical model for what we want to represent • When the linear model intercepts the x-axis, the + sign changes to a - sign despite the electric field not actually changing to the opposite direction.	• Provides a mathematical model with an **improved spatial dependence** on the electric field



⇒ many experimental challenges

KEY

- ***Cubic model***
- ***Linear Model***

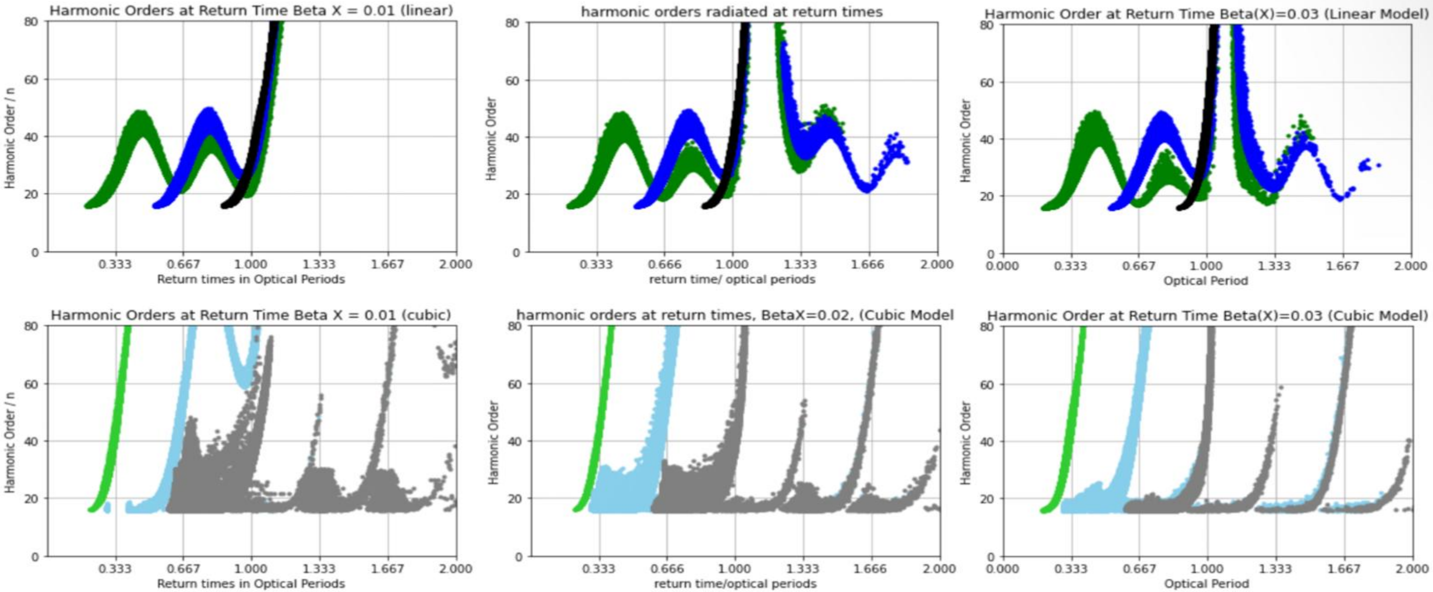
L. Ortmann and A. S. Landsman, *Physical Review A*, **97**, 023420, 2018.

Data & Results

All βx cases

linear model

Cubic model



All βy cases

linear model

Cubic model